Answer Key & Teacher Notes

Session 1 — Understanding Ratios · Session 2 — Ratios and Tape Diagrams

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Understanding Ratios

TALK PROMPTS

- Does order matter in a ratio? Is 4 : 6 the same as 6 : 4?
- Why does 4 : 3 describe the same garden as 24 : 18 (Problem 4)?

PARTNER TASK

Band Sections. In Problem 6, one partner writes trumpets to flutes in simplest form (4 : 7); the other writes drums to all band members (3 : 14). Explain why you add all the groups for a part-to-whole ratio.

WATCH FOR

- Writing the terms in the wrong order (dogs to cats written as cats to dogs).
- Using one group as the whole in a part-to-whole ratio instead of adding both groups.
- Dividing only one term when simplifying (12 : 18 written as 6 : 18).

SESSION 2 · TEACHER NOTES

Ratios and Tape Diagrams

TALK PROMPTS

- How does dividing the total (25) by the total parts (5) help you find each ingredient?
- Multiply each row by the value of one box. Do your two amounts add back to the total?

PARTNER TASK

Garden Beds. In Problem 8 (beans to corn 5 : 3, 96 seeds), one partner finds the total parts and one box (12); the other finds each amount (60 beans, 36 corn) and checks that the sum is 96.

WATCH FOR

- Dividing the total by one ratio term instead of the total parts.
- Using the unit value as the answer without multiplying by the group's number of boxes.
- Forgetting to check that the shares add back to the total.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.RP.A.1	1	2	1	2	2	2	2	2	2	1	1
S1 · Form B	6.RP.A.1	1	2	1	2	2	2	2	2	2	1	1
S2 · Form A	6.RP.A.1	2	1	1	2	2	2	2	2	2	1	1
S2 · Form B	6.RP.A.1	2	1	1	2	2	2	2	2	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

ratio: A comparison of two **amounts**.

term: One of the two **numbers** in a ratio.

part-to-part: Compares one group with **another** group.

part-to-whole: Compares one group with the **total**.

- Write a ratio in the **order** the question asks.
- For part-to-whole, **add** the groups to find the whole.
- To simplify, divide both terms by the same **number**.

Example: All books: 6 + 9 = **15**. Math first: 6 : **15**. Divide both terms by **3**: 2 : 5. Answer: Math to all books = **2** : 5

YOU TRY

1. a. 10 : 15 = 2 : 3 b. 15 : 25 = 3 : 5

2. 4 : 7

EXIT TICKET

1. a. 6 : 9 = 2 : 3 b. 9 : 15 = 3 : 5

2. 8 : 20 = 2 : 5

3. **A** (2 : 5)

PRACTICE FORM A

1. a. 4 : 6 = 2 : 3 b. 2 : 12 = 1 : 6 c. 6 : 4 = 3 : 2 d. 4 : 12 = 1 : 3

2. a. 12 to 8, 12 : 8, 12/8 = 3 : 2 b. 8 : 20 = 2 : 5

3. 15 : 10 = 3 : 2

4. 24 : 18 = 4 : 3. Yes, the helper is correct.

5. mystery = 30 − 12 = 18; 18/30 = 3/5

6. a. 8 : 14 = 4 : 7 b. 6 : 28 = 3 : 14

7. a. one box = 5, so blue = 3 × 5 = 15 marbles b. all = 5 + 15 = 20; 5 : 20 = 1 : 4

8. a. 12 : 16 = 3 : 4 b. 16 : 28 = 4 : 7 c. No. Two more votes for Ana make 14 : 16 = 7 : 8, not 1 : 1.

9. Devon reversed the order. Dogs come first, so dogs to cats is 8 to 5.

10. **B** (2 : 3)

11. **B** (5 : 12)

Self-check word **RATIOS**

PRACTICE FORM B

1. a. 6 : 10 = 3 : 5 b. 4 : 20 = 1 : 5 c. 10 : 6 = 5 : 3 d. 6 : 20 = 3 : 10

2. a. 15 to 6, 15 : 6, 15/6 = 5 : 2 b. 6 : 21 = 2 : 7

3. 14 : 6 = 7 : 3

4. 21 : 28 = 3 : 4. No, the helper reversed the order.

5. mystery = 35 − 15 = 20; 20/35 = 4/7

6. a. 6 : 10 = 3 : 5 b. 4 : 20 = 1 : 5

7. a. one box = 6, so blue = 4 × 6 = 24 marbles b. all = 6 + 24 = 30; 6 : 30 = 1 : 5

8. a. 12 : 18 = 2 : 3 b. 18 : 30 = 3 : 5 c. No. Four more votes for Ana make 16 : 18 = 8 : 9, not 1 : 1.

9. Maya reversed the order. Guinea pigs come first, so guinea pigs to rabbits is 7 to 4.

10. **D** (2 : 3)

11. **C** (4 : 13)

Self-check word **SIMPLE**

GUIDED NOTES

tape diagram: A drawing with **equal** boxes that shows a ratio.

unit (one box): The amount inside **one** box.

total parts: All the boxes **added** together.

- Add the ratio terms to find the **total** parts.
- One box = total ÷ **total parts**.
- Multiply one box by each **term** to find each share.

Example: Total parts: 3 + 4 = **7**. One box: 28 ÷ 7 = **4** stickers. Ana: 3 × 4 = **12**. Ben: 4 × 4 = 16. Answer: Ana gets **12** and Ben gets **16**. Check: 12 + 16 = 28

YOU TRY

1. a. 8 b. 9 c. 10

2. 7 parts; \$56 ÷ 7 = \$8; \$24 and \$32

EXIT TICKET

1. a. 7 b. 7 c. 8

2. 9 parts; 36 ÷ 9 = 4; 8 and 28

3. **A** (total parts)

PRACTICE FORM A

1. a. 2 + 3 = 5 parts b. 25 ÷ 5 = 5 cups c. granola 2 × 5 = 10 cups; cranberries 3 × 5 = 15 cups

2. 20 ÷ 5 = 4; bananas 2 × 4 = 8

3. 35 ÷ 5 = 7; blue 4 × 7 = 28 cups

4. 5 + 3 = 8 parts; 32 ÷ 8 = 4; goldfish 5 × 4 = 20

5. 42 ÷ 6 = 7 counselors

6. 4 + 5 = 9; \$45 ÷ 9 = \$5; \$20 and \$25

7. a. 15 ÷ 5 = 3 oz b. 7 parts × 3 = 21 oz

8. a. 96 ÷ 8 = 12; beans 60, corn 36 b. 60 − 36 = 24 c. 1 : 1 (48 and 48)

9. She divided by one term, not the total parts. Total parts = 7, one box = 28 ÷ 7 = 4, so sugar = 2 × 4 = 8 cups.

10. **B** (6)

11. **C** (20)

Self-check word **DIAGRAM**

PRACTICE FORM B

1. a. 3 + 2 = 5 parts b. 30 ÷ 5 = 6 cups c. oats 3 × 6 = 18 cups; raisins 2 × 6 = 12 cups

2. 21 ÷ 7 = 3; bananas 3 × 3 = 9

3. 40 ÷ 5 = 8; white 1 × 8 = 8 cups

4. 3 + 5 = 8 parts; 40 ÷ 8 = 5; guppies 5 × 5 = 25

5. 60 ÷ 12 = 5 counselors

6. 3 + 4 = 7; \$56 ÷ 7 = \$8; \$24 and \$32

7. a. 20 ÷ 4 = 5 oz b. 7 parts × 5 = 35 oz

8. a. 96 ÷ 12 = 8; carrots 56, peas 40 b. 56 − 40 = 16 c. 1 : 1 (48 and 48)

9. He divided by one term, not the total parts. Total parts = 7, one box = 42 ÷ 7 = 6, so rice = 3 × 6 = 18 cups.

10. **B** (4)

11. **A** (5)

Self-check word **DIVIDED**

SHEET 1 · UNDERSTANDING RATIOS

1. a. 2 : 3 b. 3 : 2 c. 6 : 15 = 2 : 5 d. 9 : 15 = 3 : 5
2. a. 2 : 3 b. 5 : 2 c. 3 : 7
3. Part-to-whole. The 15 counts all the players on the team (8 girls + 7 boys).
4. Aquarium votes: 24 − 14 = 10. Museum to aquarium = 14 : 10 = 7 : 5. For every 7 museum votes, there are 5 aquarium votes.
5. Marcus reversed the order: 2 : 5 is cats to dogs (4 : 10). Dogs to cats is 10 : 4 = 5 : 2.

SHEET 2 · RATIOS AND TAPE DIAGRAMS

1. a. 3 + 5 = 8 boxes; 48 ÷ 8 = 6 beads b. Red: 3 × 6 = 18; Blue: 5 × 6 = 30 (18 + 30 = 48)
2. a. 18 and 24 b. 12 and 54
3. a. 12 ÷ 3 = 4 b. 4 × 4 = 16 melon pieces c. 12 + 16 = 28 pieces
4. a. Theo has 7 − 2 = 5 more boxes, so 5 boxes = 35 and one box = 7 stickers b. Maya 2 × 7 = 14; Theo 7 × 7 = 49; total 14 + 49 = 63 stickers
5. He divided 45 by 5 (one term) instead of by the total parts, so his shares add to 81, not 45. Total parts = 4 + 5 = 9; 45 ÷ 9 = 5 per box; shares are 4 × 5 = 20 and 5 × 5 = 25.
6. **D** ($25)
7. **B** (8)

October 2026 conceptual additions — teacher key

Session 1 · Understanding Ratios

Blue:yellow = 4:6 = 2:3; yellow:blue = 6:4 = 3:2; blue:all = 4:10 = 2:5. Order and the meaning of the second quantity matter.

Extension: No. Adding 2 of each produces 6:8 = 3:4, different from 2:3. Equivalent ratios require the same multiplicative scale, not equal additions.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Ratios and Tape Diagrams

Five equal units; 35 ÷ 5 = 7 per unit. Shares 14 and 21; 14 + 21 = 35 and 14:21 = 2:3.

Extension: Any positive multiple of 5 works, such as 40 with shares 16 and 24; explain the shared unit value.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.